Negative logarithms in Group I Row III: the $i\epsilon$ prescription

1 What your advisor means

A logarithm of a negative number is not defined until one says from which side of the cut it is approached:

$$\log(-|u| \pm i0) = \log|u| \pm i\pi. \tag{1.1}$$

The "ϵ" is the infinitesimal that tells you the side. The "π" is what it produces. Both come from the Sokhotski–Plemelj identity

$$\frac{1}{u \pm i\epsilon} = \mathrm{PV}\frac{1}{u} \mp i\pi\,\delta(u). \tag{1.2}$$

So one may not simply replace log(negative) by $\log|\cdot|$. One has to:

1. find which denominator vanishes;
2. give it its physical $i\epsilon$;
3. keep the resulting $\mp i\pi$;
4. see what the $i\pi$ does in the cross section.

Steps 1–4 are done below. The outcome is that the $i\pi$ terms are *not* zero term by term. They combine with the c.c. into a real contribution proportional to π. That contribution vanishes after averaging over a Gaussian (C-even) projectile, which is why $\log|\cdot|$ is the right final answer. It is now derived, not assumed.

2 Step 1: which denominator vanishes

In Lublinsky–Mulian (1610.03453) the state that is dressed is the vacuum. Every energy denominator is minus a sum of positive light-cone energies, e.g. $E_{gg} = \frac{\mathbf{k}^2}{2k^+} + \frac{\mathbf{p}^2}{2p^+}$ in their (3.7)–(3.17). Such a denominator never vanishes, so all their logs have positive arguments, e.g. (C.2)–(C.6). Their complex-analysis ingredients are:

- the $2i\pi$ of the Fourier transforms (C.28)–(C.30);
- the remark below (4.21) that the complex-conjugate term comes with the opposite sign because the prefactor is if^{abc} times a real kernel. That is exactly the mechanism of Step 4.

In your Group I Row III the state is a *gluon*, $|g_i^a(k)\rangle$ (the $\mathfrak{B}_3$ wave function). Its denominators are energy *differences*, e.g. `Calculation_main.tex` Eq. (oneginst1):

$$\frac{\langle g(p)|H_{gg\,\mathrm{inst}}|g(k)\rangle}{E_g(k) - E_g(p)}, \qquad E_g(k) - E_g(p) = \frac{p^+\mathbf{k}^2 - k^+\mathbf{p}^2}{2k^+p^+}. \tag{2.1}$$

This vanishes when the intermediate gluon is on shell with the same energy as the incoming one, $p^+\mathbf{k}^2 = k^+\mathbf{p}^2$. That is a real, not virtual, transition. After the Fourier transform (i3) it becomes your denominator $p^+(x-z)^2 - k^+(x-w)^2$, which vanishes at

$$p_0^+ = k^+\frac{(x-w)^2}{(x-z)^2}. \tag{2.2}$$

3 Step 2: the $i\epsilon$, and what it does to (i3) and to the p^+ integral

The prescription. A state built from an incoming state by adiabatic switching (Lippmann–Schwinger) has $\frac{1}{E_{\mathrm{in}} - E_n + i\epsilon}$. Here this gives

$$\frac{1}{E_g(k) - E_g(p)} \;\to\; \frac{2k^+p^+}{p^+\mathbf{k}^2 - k^+\mathbf{p}^2 + i\epsilon} = \frac{2p^+}{\xi\mathbf{k}^2 - \mathbf{p}^2 + i\epsilon}, \qquad \xi = \frac{p^+}{k^+}. \tag{3.1}$$

The conjugate amplitude has $-i\epsilon$. (An outgoing state would have the opposite sign everywhere; this does not change the conclusion of Step 4.)

Fourier transform (i3) with $i\epsilon$. Use $\frac{1}{Q+i\epsilon} = -i\int_0^\infty dt\, e^{it(Q+i\epsilon)}$ with $Q = \xi\mathbf{k}^2 - \mathbf{p}^2$, and the Gaussian integrals

$$\int \frac{d^2k}{2\pi} e^{i\mathbf{k}\cdot X + it\xi\mathbf{k}^2} = \frac{i}{2t\xi} e^{-iX^2/4t\xi}, \qquad \int \frac{d^2p}{2\pi} e^{i\mathbf{p}\cdot Z - it\mathbf{p}^2} = \frac{-i}{2t} e^{iZ^2/4t}.$$

With $s = \frac{1}{4t}$:

$$\int \frac{d^2k}{2\pi}\frac{d^2p}{2\pi} \frac{e^{i\mathbf{k}\cdot X + i\mathbf{p}\cdot Z}}{\xi\mathbf{k}^2 - \mathbf{p}^2 + i\epsilon} = -\frac{i}{\xi}\int_0^\infty ds\, e^{is(Z^2 - X^2/\xi)} = \frac{1}{\xi Z^2 - X^2 + i0}. \tag{3.2}$$

This is your (i3) with its $i0$ restored; the $i\epsilon$ survives with the same sign. In your variables the p^+ integrand of every log of Row III becomes $\frac{A}{p^+A - k^+B + i0}$, with $A = (x-z)^2$ and $B = (x-w)^2$.

The p^+ integral. The antiderivative is $\log(p^+A - k^+B + i0)$. Its argument stays in the upper half plane, so it is continuous across p_0^+:

$$\boxed{\int_{p_1}^{p_2} dp^+ \frac{A}{p^+A - k^+B + i0} = \log\left|\frac{p_2A - k^+B}{p_1A - k^+B}\right| - i\pi\,\theta\big(p_1 < p_0^+ < p_2\big)} \tag{3.3}$$

The real part is the $\log|\cdot|$ you already have. The new piece is $-i\pi$ exactly where the argument was negative. Checked in `check.py`:

- $\epsilon = 10^{-6}$ gives $\mathrm{Im}/\pi = -0.9999998$ when p_0^+ is inside the range and -10^{-6} when it is outside;
- direct quadrature at $\epsilon = 10^{-4}$ gives $\mathrm{Re} = 0.98738665$ and $\mathrm{Im}/\pi = -0.99998$.

For the logs of Row III:

log in Row III	p^+ range	$-i\pi$ present iff
$\log\left\lvert\frac{(x-z)^2 - \frac{k^+}{\vee}(x-w)^2}{(x-z)^2-(x-w)^2}\right\rvert$ (part 1 l.2, part 2)	$(k^+, \vee)$	$\frac{k^+}{\vee}(x-w)^2 < (x-z)^2 < (x-w)^2$
$\log\left\lvert\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right\rvert$ (part 1 l.3, part 3a)	$(0, k^+)$	$(x-z)^2 > (x-w)^2$
$\log\left\lvert\frac{(x-z)^2 - \frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right\rvert$ (part 1 l.4)	$(0, \vee)$	$(x-z)^2 > \frac{k^+}{\vee}(x-w)^2$

Write θ_j for the corresponding step function. It is a real function of x, z, w.

4 Step 3: what the $i\pi$ does in the cross section

Every log term of Row III has the form

$$T = i\, f^{abd}\, K\, L\, \mathcal{O}, \qquad \mathcal{O} \equiv \big[U(x') - U(w')\big]^{ab'} U^{de}(x) \big[U(z) - U(y)\big]^{bc}\, \rho^{b'}(x')\rho^e(x)\rho^c(y), \tag{4.1}$$

where:

- K is a real transverse kernel and L the log;
- f and the adjoint U's are real;
- Group I enters as $T + T^\dagger$; the conjugate reverses the order of the ρ's and carries $-i0$, i.e. $+i\pi$.

With $L = \log|\cdot| - i\pi\theta$:

$$T + T^\dagger = \underbrace{i f^{abd} K\, \log|\cdot|\,\big(\mathcal{O} - \mathcal{O}^\dagger\big)}_{\text{what you have now}} + \underbrace{\pi\, f^{abd} K\, \theta\,\big(\mathcal{O} + \mathcal{O}^\dagger\big)}_{\text{new: from the } i\pi}, \tag{4.2}$$

with $\mathcal{O}^\dagger = [\dots]\,\rho^c(y)\rho^e(x)\rho^{b'}(x')$ (same c-number colour tensor, reversed order).

- The $\log|\cdot|$ multiplies the *antisymmetric* combination $\mathcal{O} - \mathcal{O}^\dagger$. This is the sign flip noted under Lublinsky–Mulian (4.21).
- The $i\pi$ multiplies the *symmetric* combination $\mathcal{O} + \mathcal{O}^\dagger$, with a real coefficient π.

So the $i\pi$ does not drop out by itself. This agrees with what I found earlier (`pole_check.py`): the residues at p_0^+ do not cancel between the terms of the row.

5 Step 4: the projectile average kills the $i\pi$ term (Gaussian projectile)

Take $\langle \mathcal{O} + \mathcal{O}^\dagger \rangle$ with the rules used for the two-ρ reduction: $[\rho^a(u), \rho^b(v)] = if^{abc}\rho^c(u)\delta(u-v)$ and a Gaussian weight.

1. **Distinct points** x', x, y: the ρ's commute, so $\mathcal{O} + \mathcal{O}^\dagger = 2\mathcal{O}$. The Gaussian average of an odd number of ρ's is zero.

2. **Coincident points** (the δ-function terms that make the two-ρ structures): reordering a three-ρ word gives

$$\rho_1\rho_2\rho_3 = S(\rho_1\rho_2\rho_3) + \tfrac{1}{2}\sum_{i<j}[\rho_i, \rho_j](\dots) + (\text{one-}\rho).$$

   For the reversed word every commutator changes sign, $[\rho_j, \rho_i] = -[\rho_i, \rho_j]$. So the two-$\rho$ terms cancel exactly in $\mathcal{O} + \mathcal{O}^\dagger$ and *double* in $\mathcal{O} - \mathcal{O}^\dagger$.

3. What is left of $\mathcal{O} + \mathcal{O}^\dagger$ is the totally symmetric part, $\propto d^{abc}\rho^a\rho^b\rho^c$, and one-$\rho$ terms. Both average to zero for a Gaussian (MV) projectile.

Therefore

$$\boxed{\left\langle \pi f^{abd} K\, \theta\, (\mathcal{O} + \mathcal{O}^\dagger)\right\rangle_{\text{Gaussian}} = 0, \qquad \left\langle T + T^\dagger \right\rangle = \left\langle if^{abd} K\, \log|\cdot|\, (\mathcal{O} - \mathcal{O}^\dagger)\right\rangle.} \tag{5.1}$$

Colour check (`check.py`): $\mathrm{Tr}(T^aT^bT^c) + \mathrm{Tr}(T^cT^bT^a)$, the reversed-order sum, is the symmetric part only.

- SU(2): it is 0 exactly, while $\mathrm{Tr}(abc) - \mathrm{Tr}(cba)$ is not.
- SU(3) fundamental: it equals $d^{abc}/2$ exactly.

So the $i\pi$ term would survive only through d^{abc}, i.e. a C-odd (odderon) three-ρ correlator.

6 What to write in the paper

1. The energy denominators of the gluon-state wave function, $E_g(k) - E_g(p)$ (and $E_g(k) - E_{gg}$), carry $+i\epsilon$. In coordinate space (i3)–(i5) become $\frac{1}{\xi Z^2 - X^2 + i0}$.

2. The p^+ integrals then give $\log|\cdot| - i\pi\theta$ (Eq. 3.3). The θ's are in the table of Step 2.

3. In the cross section (term + c.c.) the $\log|\cdot|$ multiplies $\mathcal{O} - \mathcal{O}^\dagger$ and the $i\pi$ multiplies $\pi(\mathcal{O} + \mathcal{O}^\dagger)$, Eq. (4.2).

4. For a Gaussian projectile $\langle \mathcal{O} + \mathcal{O}^\dagger \rangle = 0$, so only $\log|\cdot|$ remains. Beyond the Gaussian, the $i\pi$ terms give a C-odd contribution:

$$\pi f^{abd} K\, \theta\, \langle d\text{-part of } \rho\rho\rho\rangle.$$

Nothing in the evolution (BFKL/JIMWLK), DGLAP or running-coupling results changes: they come from $\mathcal{O} - \mathcal{O}^\dagger$ and from logs whose arguments are positive.
Correction to my earlier note: I justified $\log|\cdot|$ by saying (i3)–(i5) are principal values. The correct reason is the one above. The $i\epsilon$ gives $-i\pi\theta$, and that term is removed by the projectile average, not by the prescription.